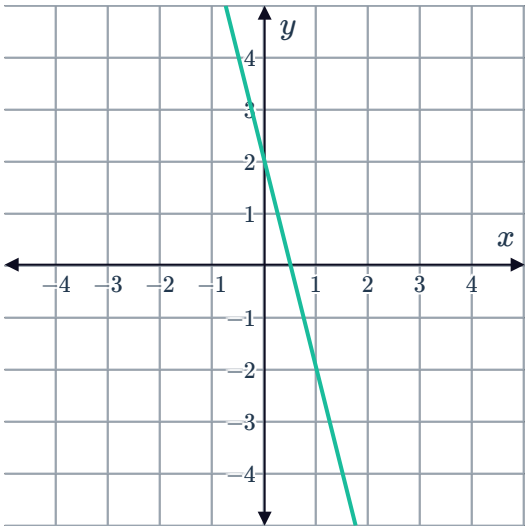
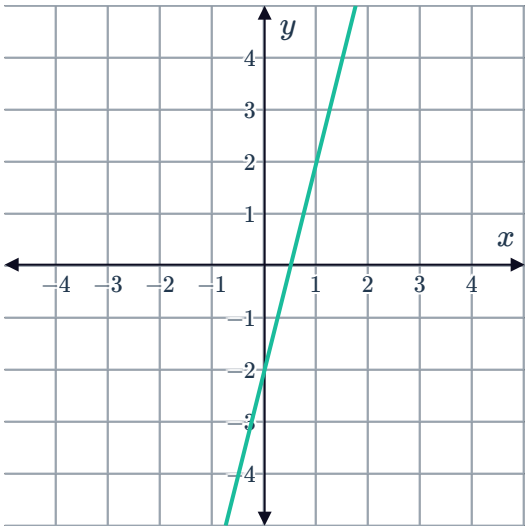
Transformations of y=x

1 Determine whether the following graphs are decreasing linear function.

a



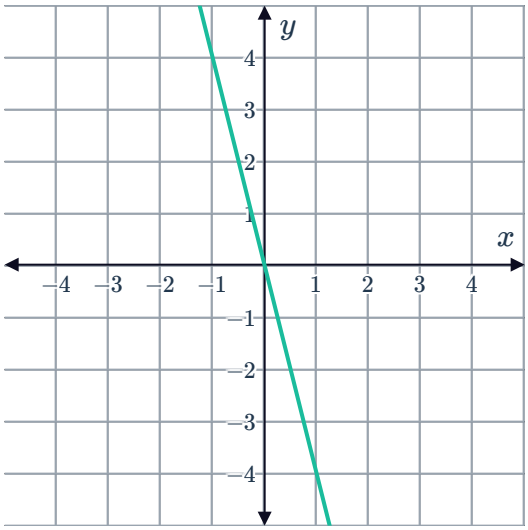
b



2 Consider the graph of $y = -4x$.

a As x becomes larger in the positive direction (i.e.x approaches ∞), what happens to the corresponding y-values?

b As x becomes larger in the negative direction (i.e.x approaches $-\infty$), what happens to the corresponding y-values?

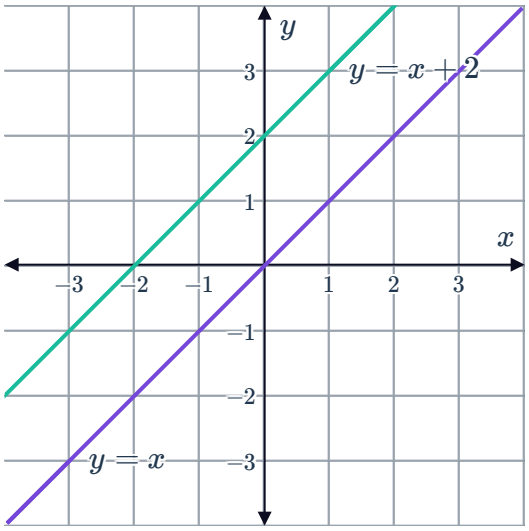


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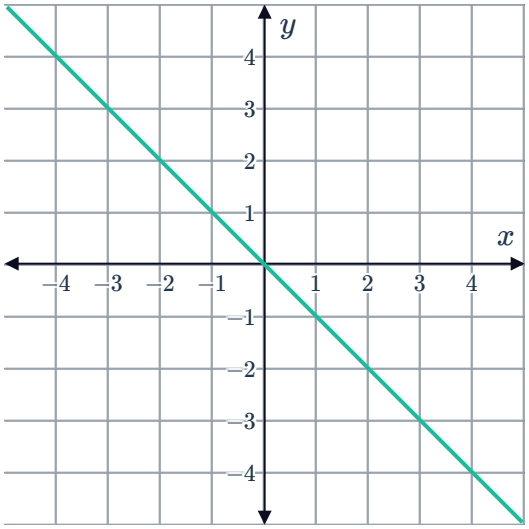
3 The lines $y = x + 2$ and $y = x$ have been graphed on the same coordinate plane. Determine whether the line $y = x + 2$ is:

a Increasing or decreasing.

b Steeper, flatter or equally steep as $y = x$.

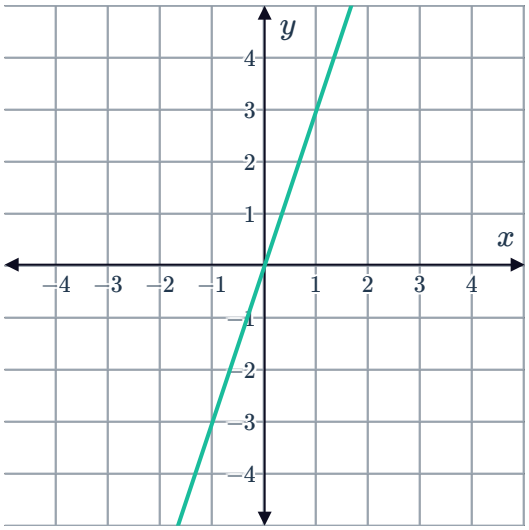


4 Consider the given graph of $y = -x$.

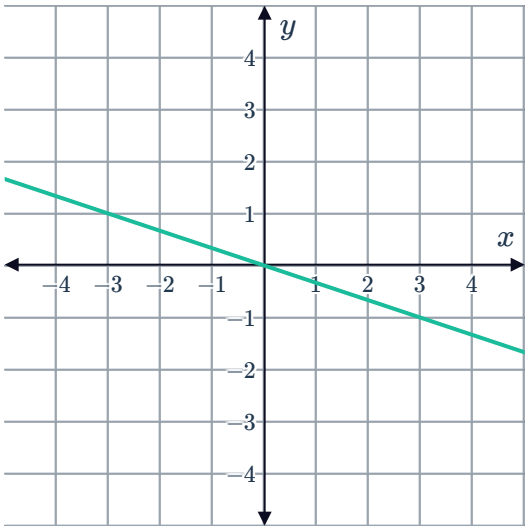


Determine whether the following shows the graph of a straight line whose y value is decreasing at a faster rate than $y = -x$.

a

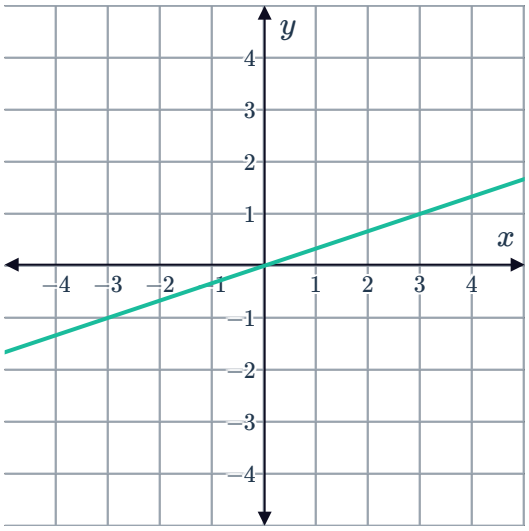


b

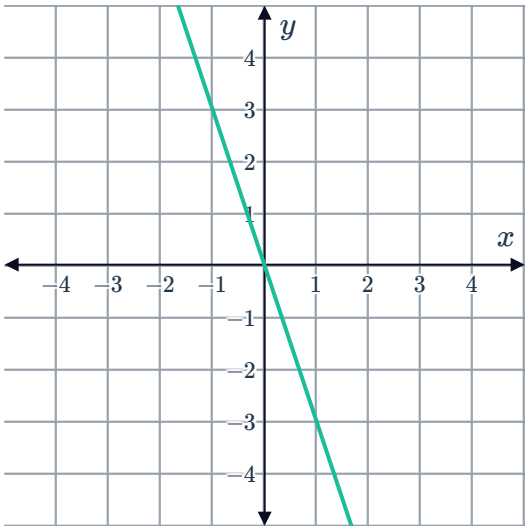


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c



d



5 A straight line graph has a positive y-intercept and a positive x-intercept. Determine whether the following must also be true of the straight line graph.

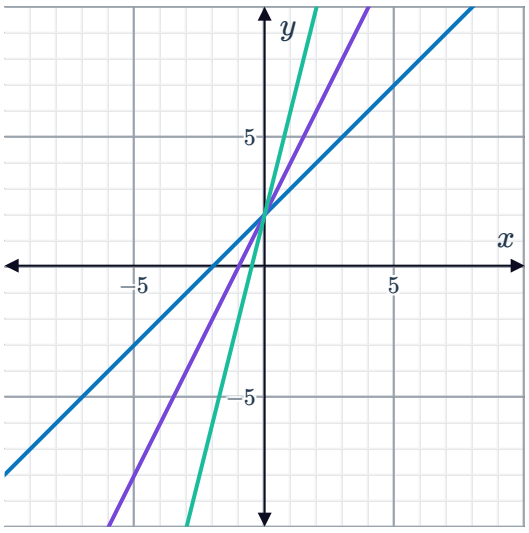
a The straight line graph is flatter than $y = x$.

b The straight line graph is increasing.

c The straight line graph is steeper than $y = x$.

d The straight line graph is decreasing.

6 The equations: $y = x + 2$, $y = 2x + 2$ and $y = 4x + 2$ have been graphed on the same number plane:



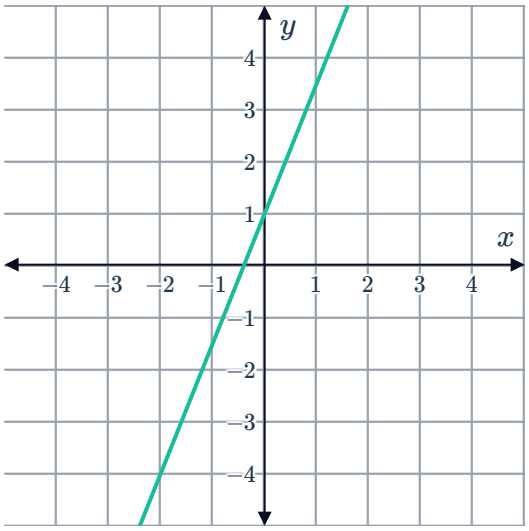
a What do all of the equations have in common?

b What do all of the graphs have in common?

c What can you conclude from the answers above?

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7 Look at the graph of the line.



a At the y-intercept, $x = 0$.
What is the y-value at this point?

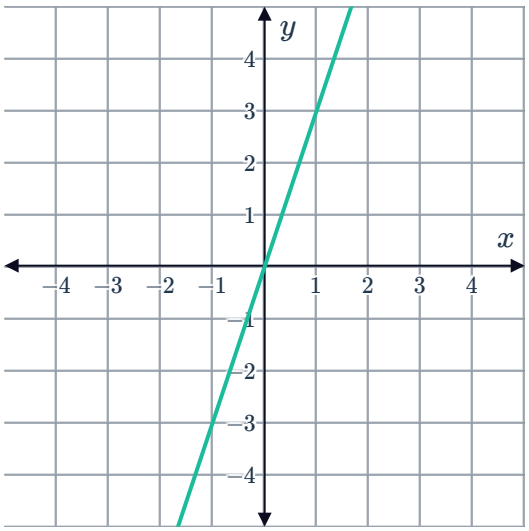
b The equation of the line is $y = \frac{5x}{2} + 1$.
For a straight line equation of the form $y = mx + b$, which variable represents the y-intercept?

8 Consider the following graphs:

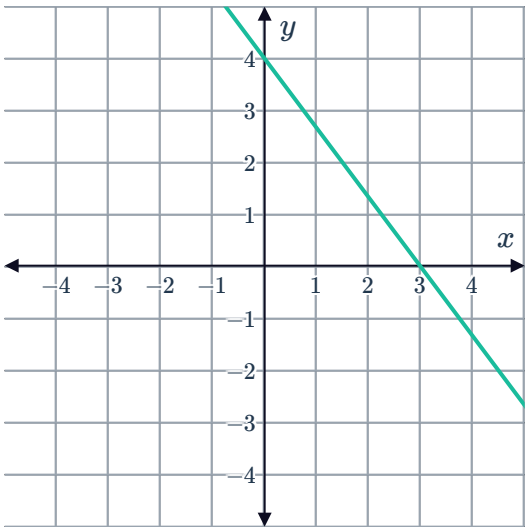
i State the x-value of the x-intercept.

ii State the y-value of the y-intercept.

a

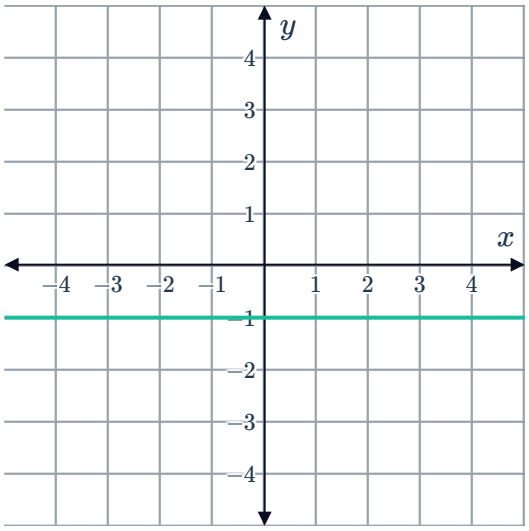


b



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c



9 Determine whether the following equations represent lines that will intersect the y-axis at 2.

a $y = 2x - 5$ **b** $y = 2x$

c $y = 2 - x$ **d** $y = 5x + 2$

e $y = x - 2$ **f** $y = x + 2$

10 A linear graph has a negative y-intercept and a positive slope.
Is the x-intercept negative or positive?

11 Determine whether the following lines pass through the origin.

a $y = 6x$ **b** $y = \frac{x}{6}$ **c** $y = 6x - 6$ **d** $y = 8x + 8$

e $y = 8x$

12 Find the y-value of the y-intercept for the following.

a $y = -7x + 5$ **b** $y = 5x$ **c** $y = 10 - 5x$ **d** $y = -x + 1$

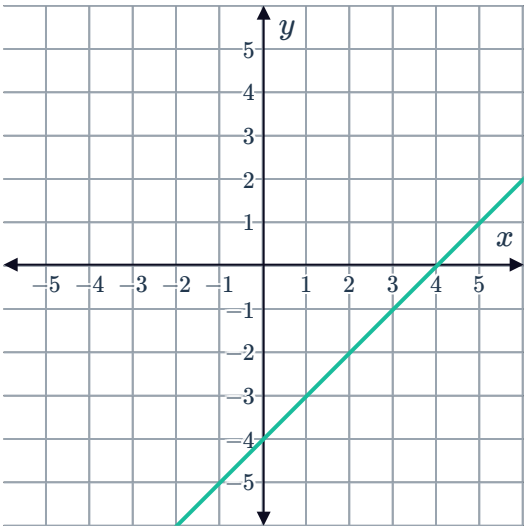
13 Determine whether the following lines have a y-intercept.

a $y = x$ **b** $x = 3$ **c** $y = -3$ **d** $3x - y = 0$

e $y = 4x + 5$

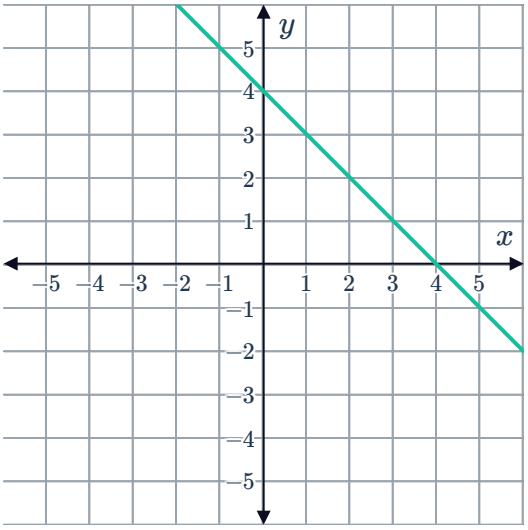
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14 Consider the graph of $y = x - 4$.



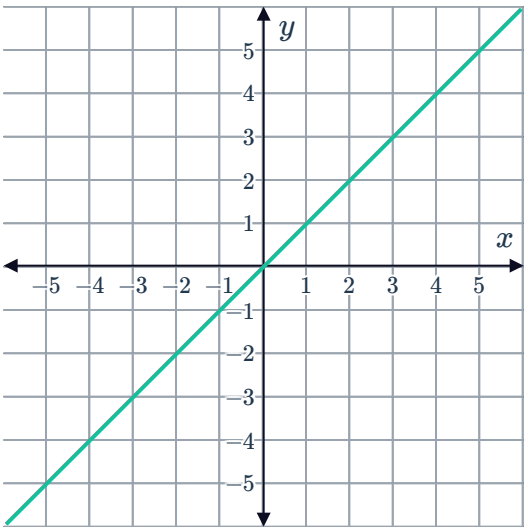
- **a** State the coordinates of the x-intercept.
- **b** State the coordinates of the y-intercept.
- **c** If the graph is translated 6 units down, find the coordinates of the new y-intercept.

15 Consider the graph of $y = 4 - x$.



- **a** State the coordinates of the x-intercept.
- **b** State the coordinates of the y-intercept.
- **c** If the graph is translated 10 units up, find the coordinates of the new y-intercept.

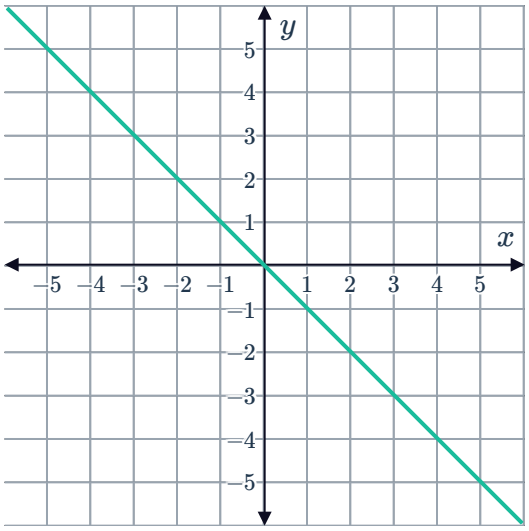
16 Consider the graph of $y = x$. Draw the graph of the resulting line translated 2 units down.



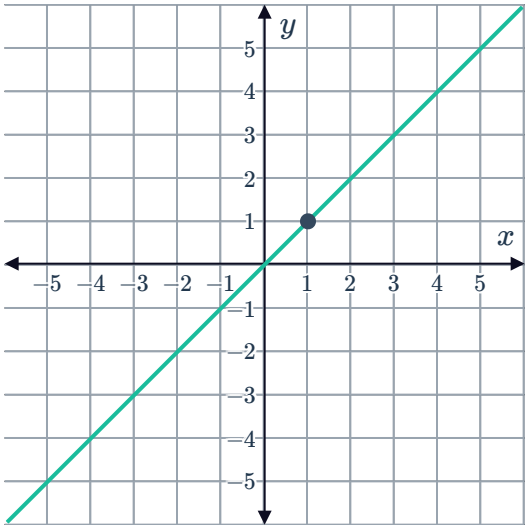
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17 Consider the graph of $y = -x$. Draw the graph of the resulting line translated 4 units down.

.



18 Consider the graph of $y = x$, with the point $(1, 1)$ that lies on the graph. The steepness of this graph is to be quadrupled.
Draw the graph of the resulting line.



19 Part of a continuous straight line graph has been graphed.
Determine whether each of the following statements is true about the entire line graph.

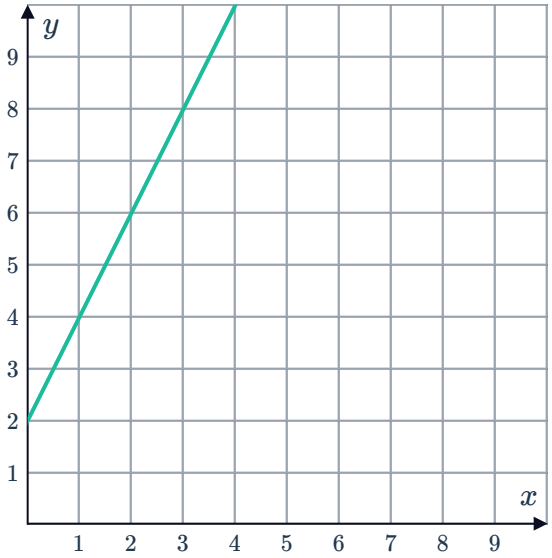
a The points on the graph have both positive and negative y values.

b The graph has a negative x-intercept.

c The graph passes through the origin.

d The points on the graph have only positive y values.

e It has a negative slope.



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20 Part of a straight line graph has been graphed.
Determine whether each of the following statements is true of the entire line graph.

a The points on the graph have both positive and negative y values.

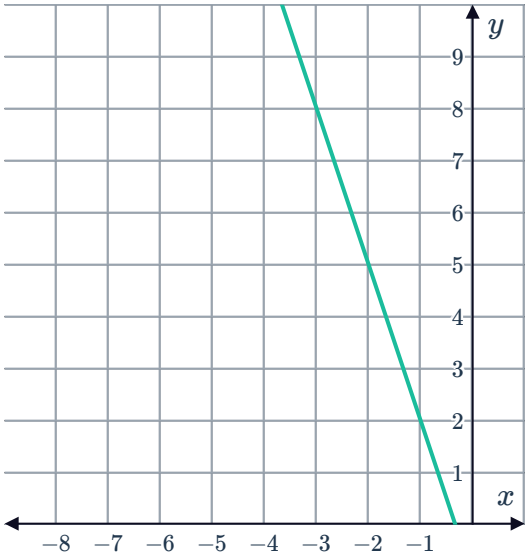
b The points on the graph have only negative y values.

c It has a negative slope.

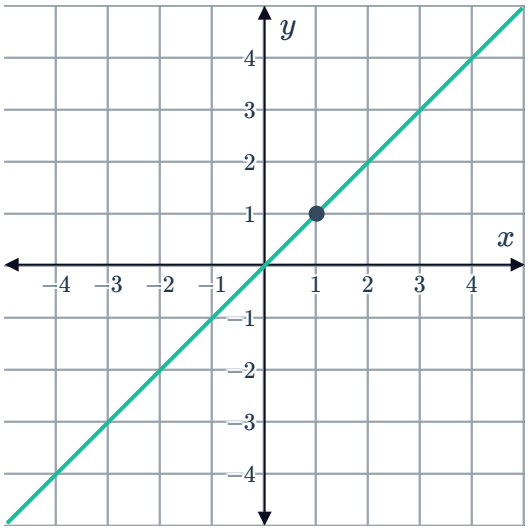
d The graph is increasing.

e The graph has a positive y-intercept.

f The graph passes through the origin.

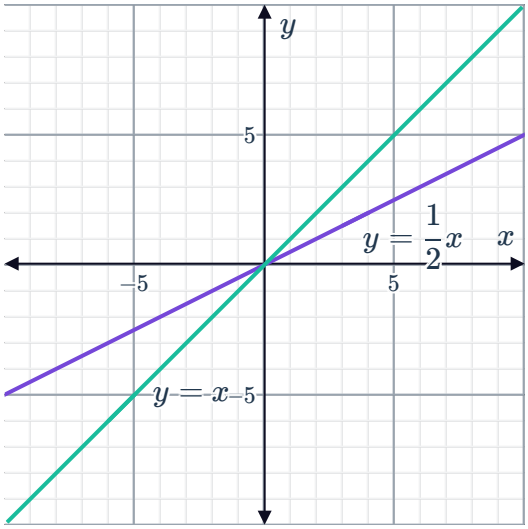


21 Consider the graph of $y = x$.
Where will the point $(1, 1)$ move to if the line is made 3 times as steep?



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22 The lines $y = \frac{1}{2}x$ and $y = x$ have been graphed on the same set of axes.



a Is the line $y = \frac{1}{2}x$ increasing or decreasing?

b Is it steeper, flatter or equally steep as $y = x$?

c Determine whether the following could be done to transform the graph of $y = x$ into $y = \frac{1}{2}x$.

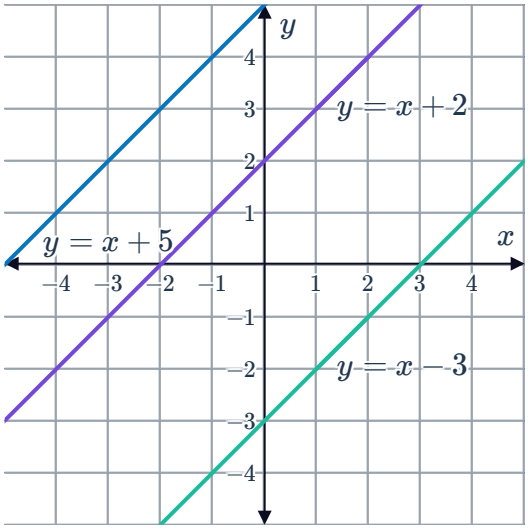
i Reflecting $y = x$ about the x-axis.

ii Reflecting $y = x$ about the y-axis.

iii Making $y = x$ steeper.

iv Making $y = x$ flatter.

23 Consider the graph of the lines $y = x + 5$, $y = x + 2$ and $y = x - 3$.



a Determine whether the following statements are true.

i They are all the same steepness.

ii All lines that are parallel to $y = x$ have positive y-intercepts.

iii All lines that are parallel to $y = x$ have positive x-intercepts.

iv They can all be made to overlap $y = x$ through a vertical translation.

v They are all parallel to one another.

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b Determine whether the following are true about the x- and y-intercepts of lines that are parallel to $y = x$.

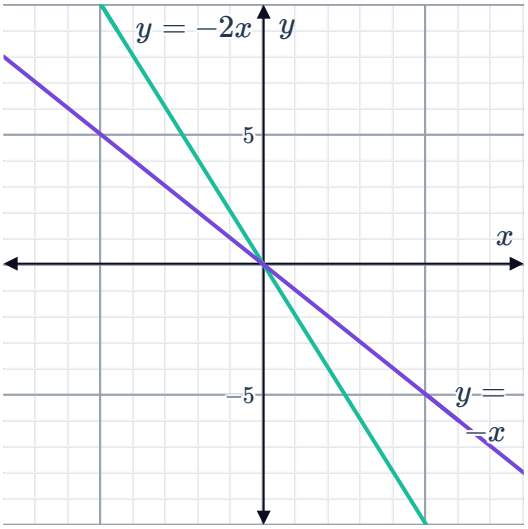
i Each line has x and y intercepts that are equal.

ii The x and y intercepts of these lines are the same distance from the origin.

iii Each line has x and y intercepts that are equal in magnitude, but opposite in sign.

c If a straight line is parallel to $y = x$ and has a y-intercept of -8, find its x-intercept.

24 The lines $y = -2x$ and $y = -x$ have been graphed on the same set of axes.



a Is the line $y = -2x$ increasing or decreasing?

b Is it steeper, flatter or equally steep as $y = -x$?

c Here are some transformations.

1. Reflection about the x-axis.
2. Steepening of straight line.
3. Reflection about the y-axis.
4. Translation 2 units down.
5. Flattening of straight line.
6. 90° rotation about the origin.

Determine whether the following pairs of transformations turn the graph of $y = x$ into the graph of $y = -2x$.

i Transformations 1 and 2

ii Transformations 2 and 3

iii Transformations 1 and 5

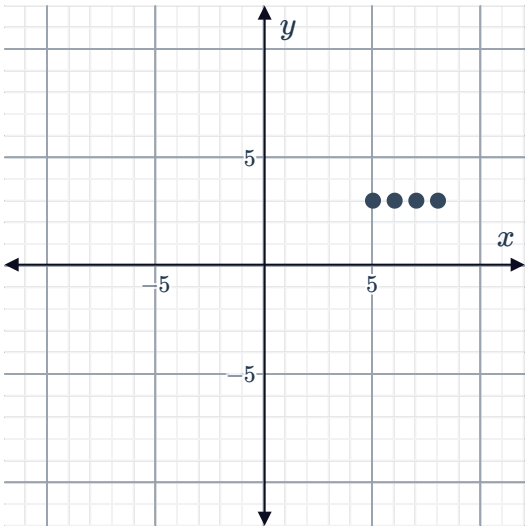
iv Transformations 2 and 6

v Transformations 4 and 5

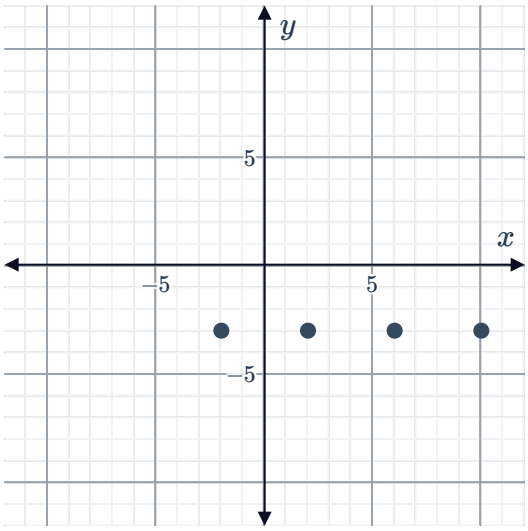
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25 Which of the following statements is true for each of the following set of points.
Statement 1: The set of points lie on a vertical line.
Statement 2: The set of points lie on a horizontal line.
Statement 3: The set of points lie on an increasing line.
Statement 4: The set of points lie on a decreasing line.

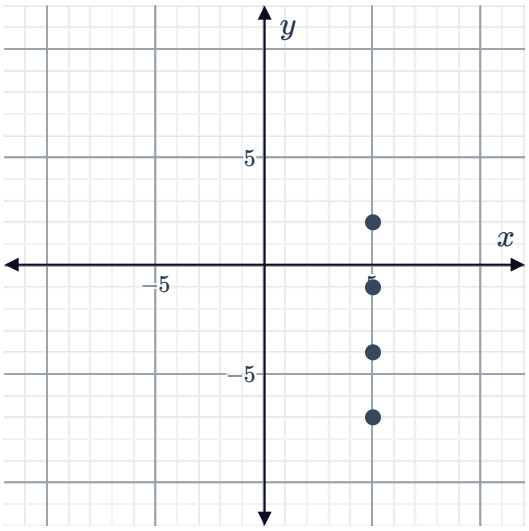
a



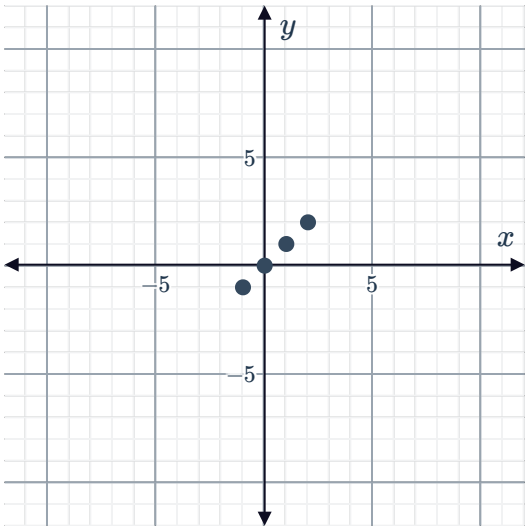
b



c

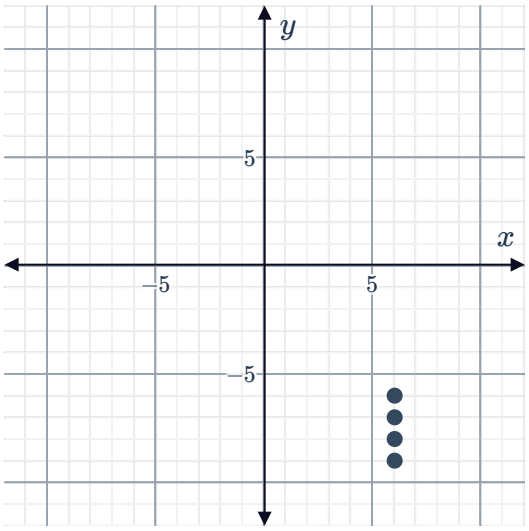


d

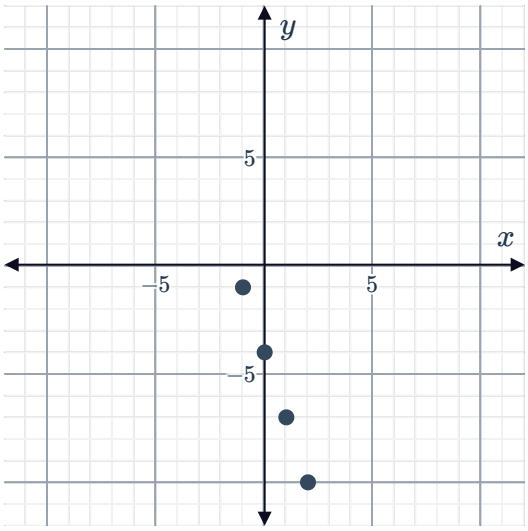


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e



f



26 Determine whether the graph of the following are horizontal or vertical line.

a $y = 6$

b $x = -4$

27 Plot the graph of the following lines:

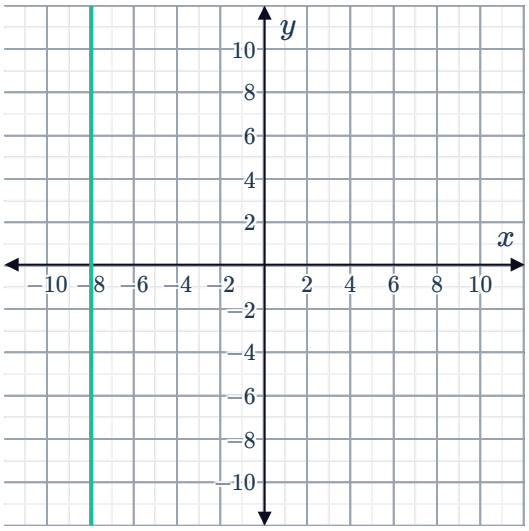
a $y = -7$

b $x = -8$

28 Consider the graphed line.

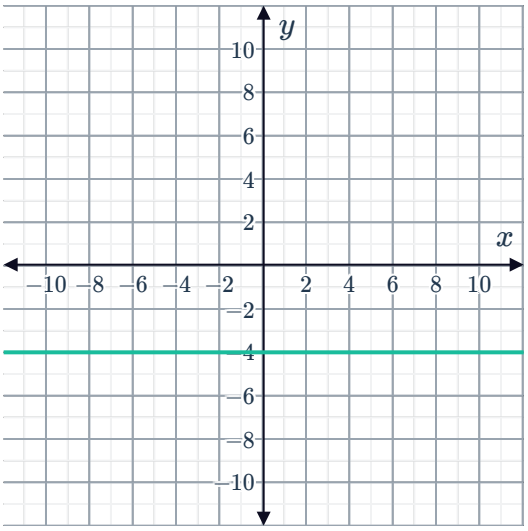
a State the x-value of the x-intercept of the line.

b How many y-intercepts does the line have?



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29 Consider the graphed line.

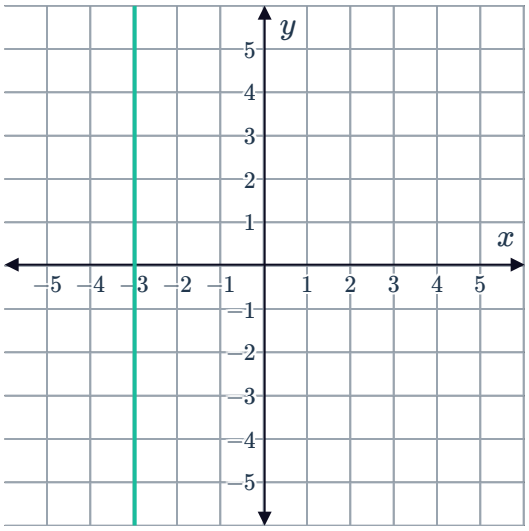


a Determine the y-value of the y-intercept of the line.

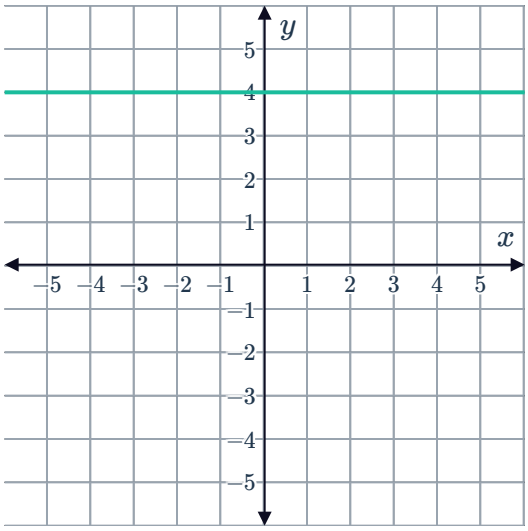
b How many x-intercepts does the line have?

30 Determine the equation of the following graphed lines.

a



b



31 Consider the table of values shown.
Find the equation of the line which passes through these points.

x	-4	-2	0	2
y	-4.5	-4.5	-4.5	-4.5

32 The table shows some points on the line $y = 0$.
The line $y = 0$ represents which axis?

x	-4.5	-0.5	3.5	8.5
y	0	0	0	0

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33 The table shows some points on the line $x = 0$.
The line $x = 0$ represents which axis?

x	0	0	0	0
y	-5.5	-2.5	3.5	6.5

34 Find the equation for the following:

a All horizontal lines that are 10 units away from the line $y = 3$

b All vertical lines that are 7 units away from the line $x = -2$

35 Three of the lines that enclose a square on a number plane have equations $x = 0$, $x = 2$ and $y = -2$.
Find all the possible equations of the fourth line.

36 The cost of a taxi ride C, in dollars, is given by $C = 2.5t + 2$, where t i duration of trip in minutes.

a For every extra minute the trip takes, how much more will the trip cost?

b State what the constant value 2 could represent.

Additional questions

37 Describe the transformation of $f(x)$ that is of the form:

a $g(x) = f(x) + k$ **b** $g(x) = kf(x)$ **c** $g(x) = -f(x)$

38 For the what values of k will the transformation from $f(x)$ to $g(x) = kf(x)$ involve a reflection across the x-axis?

39 For each of the following, describe the type of transformation from $f(x)$ to $g(x)$.

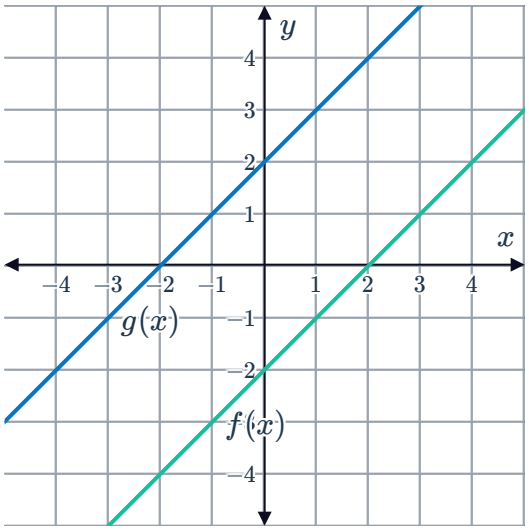
a $g(x) = f(x) + 4$ **b** $g(x) = 6f(x)$ **c** $g(x) = \frac{1}{3}f(x)$ **d** $g(x) = \frac{5}{12}f(x)$

e $g(x) = -f(x)$ **f** $g(x) = f(x) - 2$

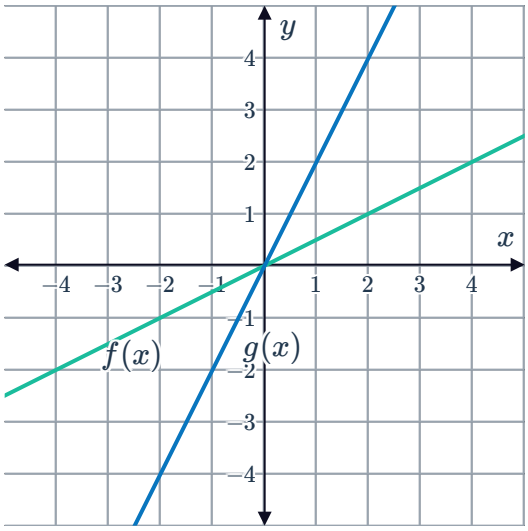
40 For each of the graphs, describe the type of transformation from $f(x)$ to $g(x)$.

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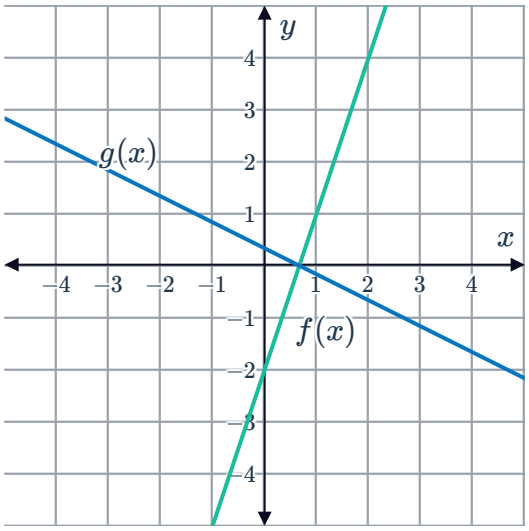
a



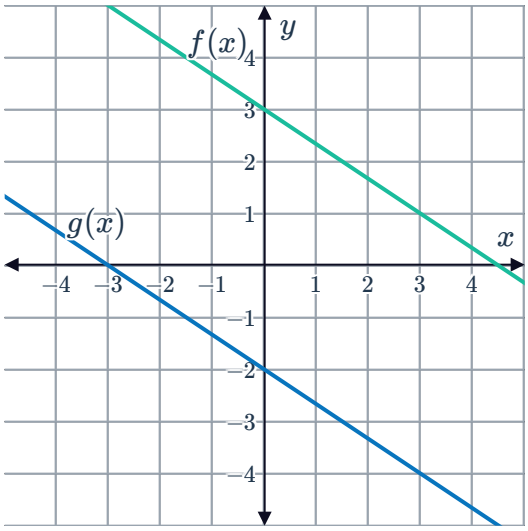
b



c

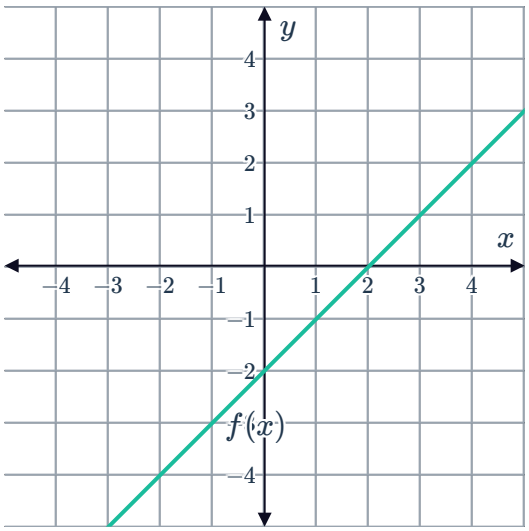


d

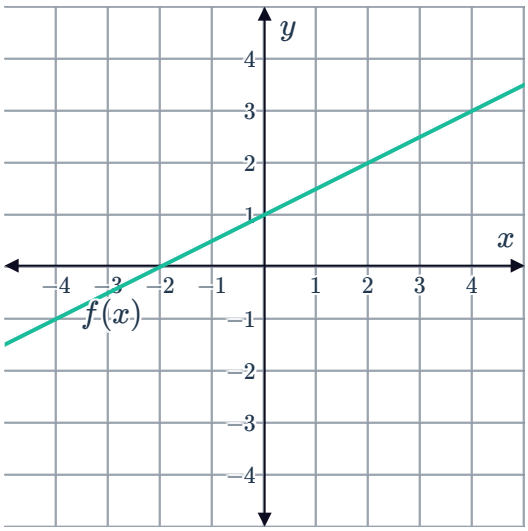


41 For each graph of $f(x)$, draw the graph of $g(x)$ using the given transformation.

a $g(x) = f(x) + 5$

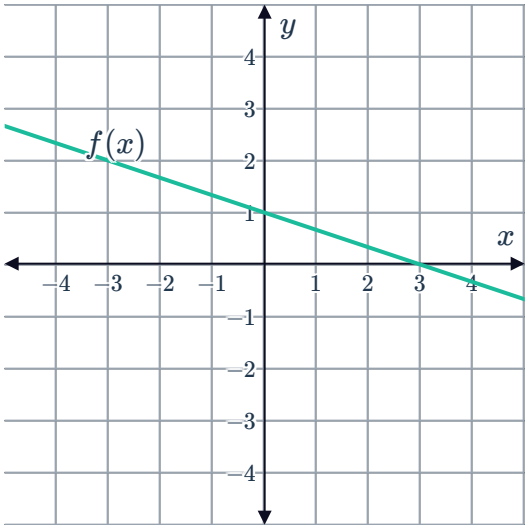


b $g(x) = 2f(x)$

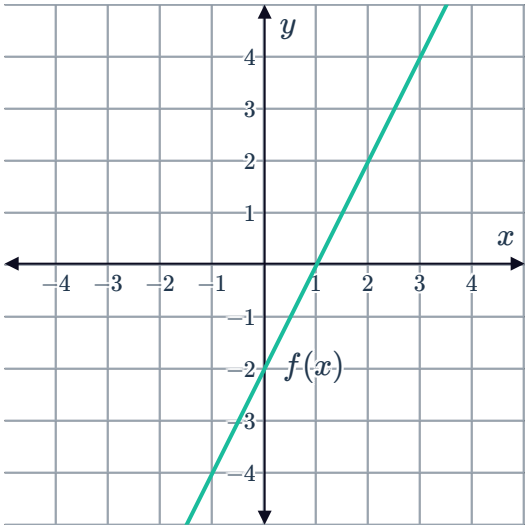


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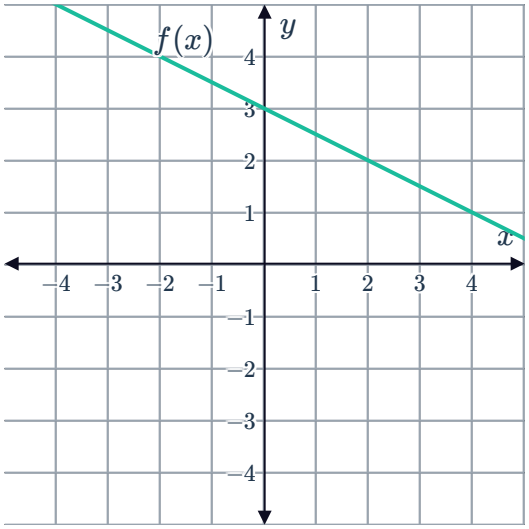
c $g(x) = -f(x)$



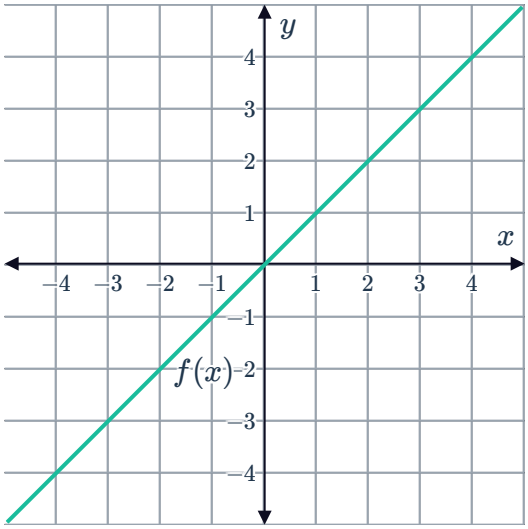
d $g(x) = -\frac{1}{4}f(x)$



e $g(x) = f(x) - 2$



f $g(x) = f(x) + 2$

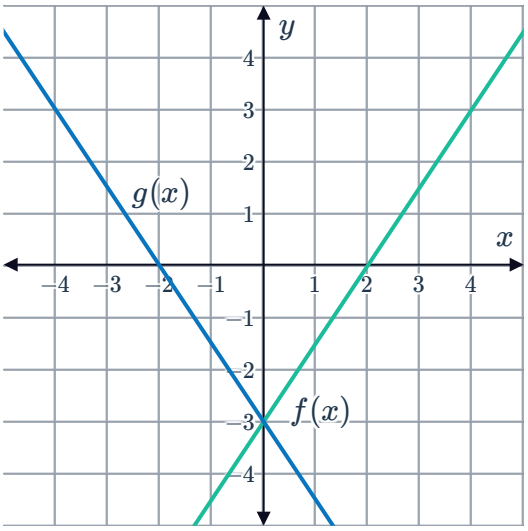


42 For each transformation, describe how the graph of $g(x)$ is related to the graph of $f(x)$.

a $g(x) = f(x) - 8$ **b** $g(x) = f(x) + \frac{1}{2}$ **c** $g(x) = -3f(x)$ **d** $g(x) = \frac{1}{4}f(x)$

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43 Oreste draws the graph of $g(x) = -f(x)$, as shown.



a Describe the error Oreste made.

b Describe the type of graphs that $f(x)$ could be for Oreste's error to still give him the correct answer.

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